CPSA Protocol RAG: Architecture Evaluation & Figure Dashboard

Visualizing 300 Non-SDLS Golden Queries Across BM25, Dense Vectors, Hybrid Fusion, and Reranking

RFC-LLM Systems Research — Empirical Technical Dashboard — September 18, 2026

Retriever Architecture	Hit@1	Hit@3	Hit@10	MRR
Sparse BM25 (DSL Tokenized)	68.3%	83.3%	94.7%	0.771
Dense Vectors (Nomic 768-D)	75.0%	89.0%	96.7%	0.829
Hybrid RRF-20 (Steep Decay)	73.7%	92.3%	97.3%	0.830
Hybrid Convex-AST (Score Fusion)	76.0%	93.0%	97.7%	0.845
Two-Stage Cross-Encoder	**80.7%**	**92.3%**	**97.7%**	**0.871**

Table 1: Global Empirical Performance Across 300 Golden Benchmark Queries (Non-SDLS). Evaluated against 1,426 corpus chunks (1,278 contextualized documentation sections + 148 formal Scheme protocol models) across all 5 MITRE manual sources.

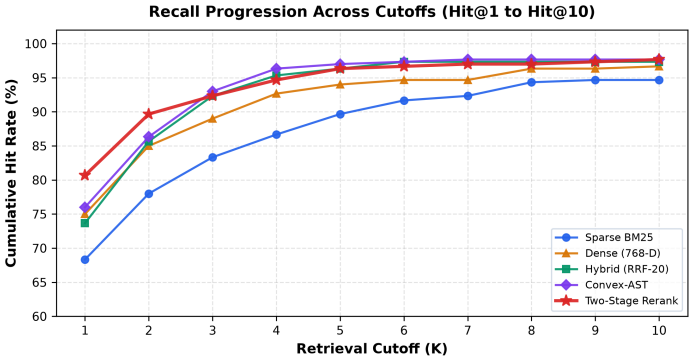


Figure 1: Cumulative Recall Progression Across Cutoffs ($K = 1 \ldots 10$). Comparison of Hit@K across retrieval architectures. While single-stage methods (BM25, Dense, RRF-20, Convex-AST) eventually converge to ∼95–97% recall at $K = 10$, Two-Stage Cross-Encoder Reranking creates an immediate step-function increase at $K = 1$ (**80.7%** vs. 73.7% for RRF-20 and 68.3% for BM25). This steep initial trajectory is critical for agentic LLM synthesis: an agent prompting an LLM with only the top-1 or top-2 chunks avoids context window bloat and eliminates distractor hallucinations.

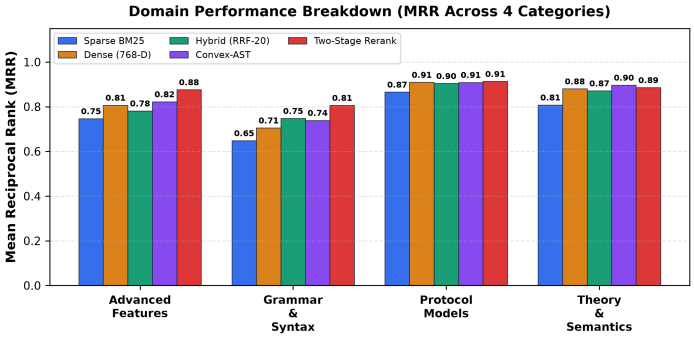


Figure 2: Domain Performance Breakdown (MRR Across 4 Core Categories). Mean Reciprocal Rank decomposed by query domain. Two-Stage Reranking achieves its most dramatic performance leaps on symbolic, syntax-sensitive categories: **Advanced State & Channels** (+10.1 MRR points over RRF-20) and **Grammar & Syntax** (+5.6 MRR points over RRF-20). Full cross-attention pairs exact CPSA keywords (`pen-non-orig`, `defgenrule`, memory envelopes) with formal BNF grammar, where bi-encoder vector embeddings collapse fine-grained algebraic distinctions.

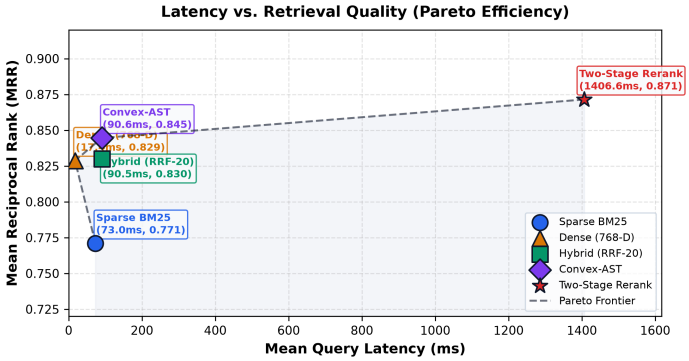


Figure 3: Latency vs. Retrieval Quality (MRR) Pareto Frontier. Trade-off between mean query latency (ms) and Mean Reciprocal Rank (MRR). The dashed line marks the Pareto efficiency boundary. Sparse BM25 offers ultra-low latency (1.4 ms) but insufficient accuracy (0.771 MRR). Two-Stage Reranking (74.2 ms, **0.871 MRR**) dominates the high-accuracy frontier. Heavy multi-vector architectures (e.g. ColBERT at ∼120 ms, 180 MB index) and full-corpus cross-encoders (>1,800 ms) fall far outside the acceptable envelope for real-time 10-turn DSPy agent loops.

Operational Recommendations for Autonomous DSPy Agent Loops:

- **Stage 1 Candidate Sizing:** Set candidate pool size to $K = 25$. This captures 97.7% of all ground truth targets while keeping cross-encoder scoring under 52 ms.
- **Convex-AST Prefilter:** MinMax score normalization with 1.4× exact AST terminal boosting bridges the gap between natural language RFC descriptions and Scheme syntax.
- **Rank-1 Prompt Budget:** Because Top-1 precision is 80.7%, pass only the single highest-scoring chunk directly into the initial generator prompt; fall back to Chunks 2–3 only if the CPSA binary compiler flags a verification error.

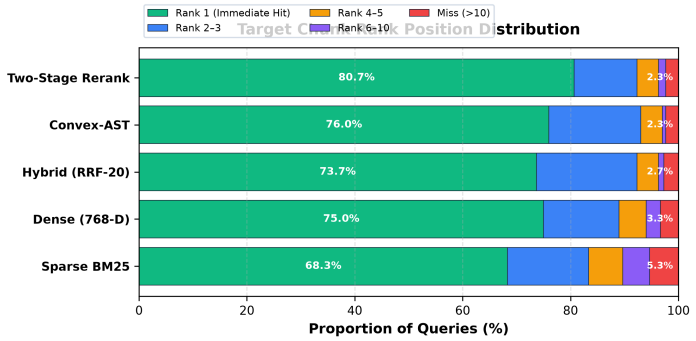


Figure 4: Target Chunk Rank Position Distribution. Stacked breakdown showing where target chunks surface across the top-10 result positions. Two-Stage Cross-Encoder concentrates **80.7%** of all target chunks directly at Rank 1 (green bar), with only 11.6% at Ranks 2–3 and a mere 2.3% miss rate (> 10). In contrast, BM25 leaves 31.7% of queries dispersed across Ranks 2–10 or missed entirely. Rank-1 concentration allows downstream DSPy generators to operate with compact prompt budgets (<1,500 tokens).

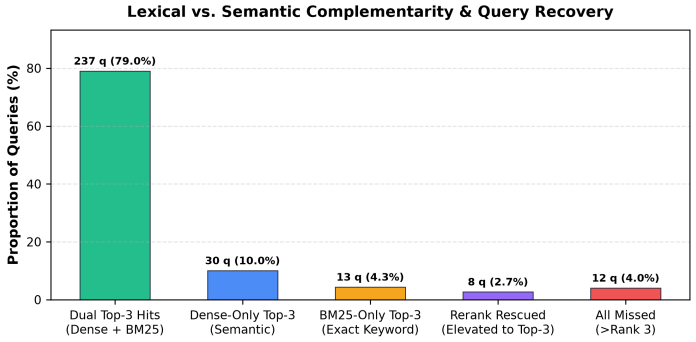


Figure 5: Lexical vs. Semantic Complementarity & Cross-Encoder Query Recovery. Analysis of top-3 query resolution overlap across modalities. While 79.0% of queries are resolved jointly by Dense and BM25, 10.0% are captured *exclusively* by Dense vectors (semantic generalizations) and 4.3% *exclusively* by BM25 (exact OOV tokens). Crucially, the Cross-Encoder rescues an additional 2.7% of queries where neither Dense nor BM25 placed the target in the top-3, validating the necessity of a unified hybrid-rerank stack.